Indirect Proofs

1. Contradiction We assume the opposite of what is being claimed and we show that assumption is false. Assume the premis but not the conclusions

Lets work with $p \iff q$

The process is:
1. assume the negation is true. $\neg p \iff q$
Recall that $\neg p \iff q \equiv \neg(p \vee q)$
Also recall that we can apply DeMorgans law to this logical equivilancy to get:
$\neg(p \vee q) \equiv p \wedge \neg q$

Applying this to "If n^2 is even then n is even" we get
n^2 is even and q is odd

Claim: "No integer is even *and* odd"

Assuming the opposite: $\exists n \in Z, n$ is even and odd

Claim: Consequitive integers have opposite parity. // Supposing the opposite, we have to check both cases, one for even and another case for odd. Formally this we're working with $/existsn, n+1$ has the same parity

fuck it here is another unfinished definiton:
$\frac{a}{b} \iff \exists k, ak = b$